

# Soldering Guide – SMD Dice

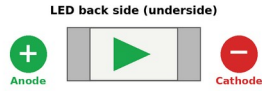
MakerMotion

Order: **smallest/flattest components first**, THT components (switch, sensor) go last.

## Tools

- Fine-tip soldering iron (320–350 °C, lead-free) · 0.5 mm solder wire · flux pen · anti-magnetic tweezers · solder wick · magnifier/microscope · multimeter

## Soldering order

| # | Component                                  | Note                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>U1 – ATtiny13A-SSU (SOIC-8)</b>         | Tin one pad, align IC (watch pin 1!), tack down, solder remaining pins. Check for bridges, ESD care.                                                                                                                                                                                                                                                                                                                                                             |
| 2 | <b>C1 – 10 µF (0805)</b>                   | Standard soldering.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3 | <b>C5 – 100 nF (1206)</b>                  | Standard soldering.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4 | <b>R2, R3, R4, R9 – 600 Ω (0805)</b>       | All four identical, order doesn't matter.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5 | <b>LED1–LED7 (1206, yellow)</b>            | Watch polarity! There's a green triangle printed on the back (underside) of the LED – flip it over and check before placing: the wide side (base) points to the plus side (anode), the tip points to the minus side (cathode) – compare with the assembly diagram. Solder one LED first, test it with a multimeter (diode test/continuity), then solder the remaining six.<br> |
| 6 | <b>BT1 – CR2032 holder (SMD)</b>           | Larger pad, needs a bit more heat/solder.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7 | <b>SW2 – tilt switch (THT)</b>             | Through-hole part.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8 | <b>U3 – SW-460D vibration sensor (THT)</b> | Through-hole part, last. Not marked, because orientation doesn't matter – the sensor has no polarity and works in any orientation.                                                                                                                                                                                                                                                                                                                               |

## Test before assembly

- Visual check for solder bridges (IC, closely spaced resistors)
- Multimeter: check continuity between adjacent IC pins
- After the first LED: diode test/continuity check with a multimeter (it will usually glow faintly) – only then solder the remaining LEDs
- Insert battery, shake → LEDs should display a number

## Troubleshooting

| Problem                       | Likely cause                                         |
|-------------------------------|------------------------------------------------------|
| No LED lights up              | Check battery polarity, check IC orientation (pin 1) |
| A single LED doesn't light up | Check that LED's solder joint/polarity               |
| Doesn't react to shaking      | Check the SW2/U3 connection to the IC                |

## Assembly diagrams (from Gerber data)

Cathode (–) red, anode (+) green – read directly from the silkscreen layer. U1 pin 1 (green dot) and BT1 polarity are confirmed against the CAD view.

